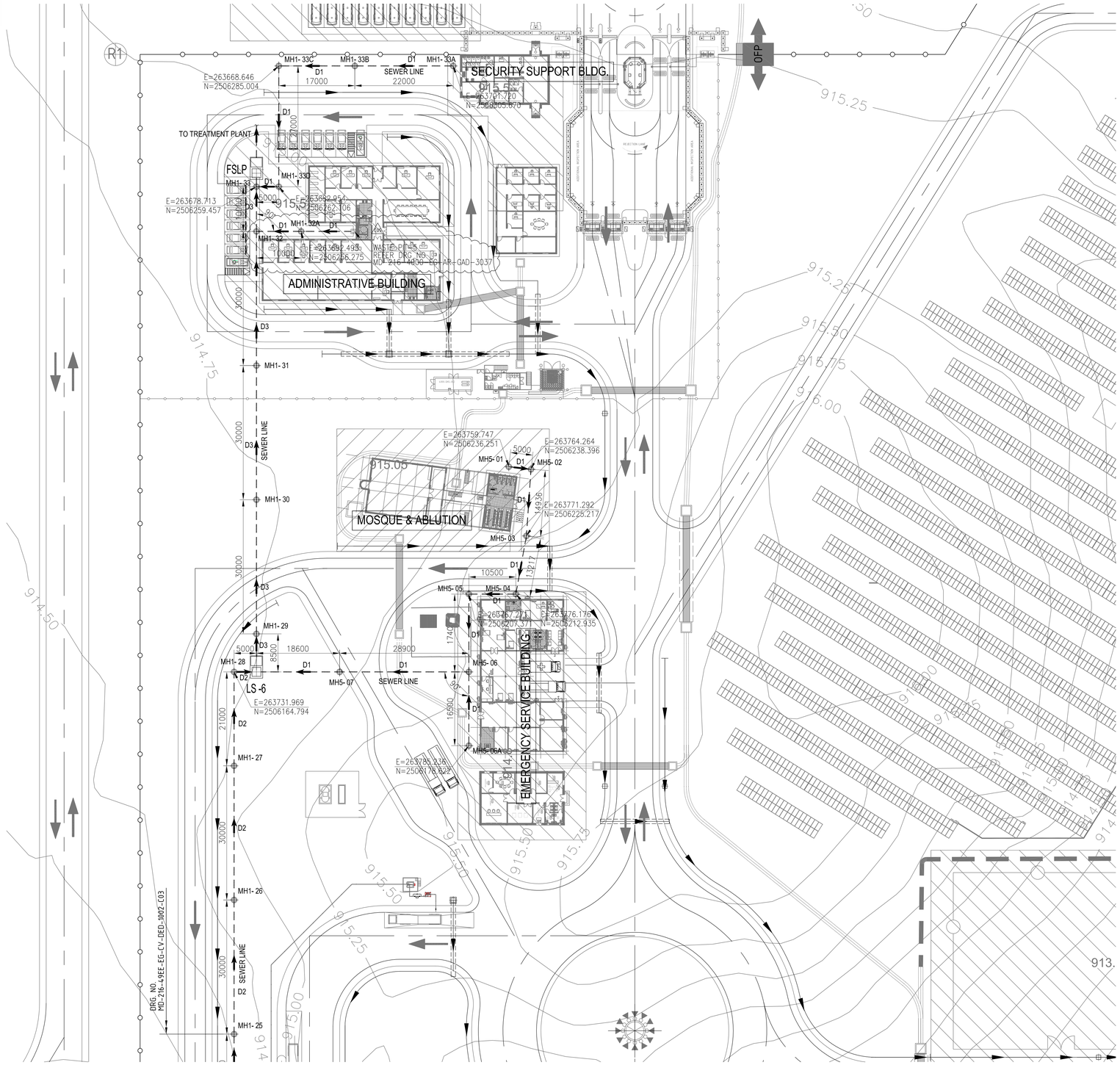
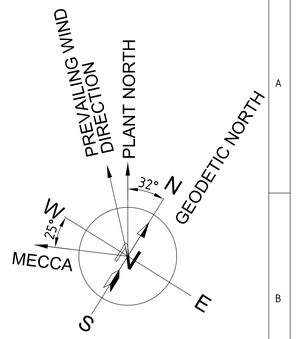
DRAWING NUMBER: MD-216-49EE-EG-CV-DED-1003
REV: D01




## LEGENDS: –

LS = LIFTING STATION
FLSP = FINAL SEWERAGE LIFTING PIT
EGL = EXISTING GROUND LEVEL
FGL = FINISHED GROUND LEVEL
DRG. = DRAWING
S.P. = SOIL PIT
W.P. = WASTE PIT

## NOTES:~

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND INVERT LEVEL & CO-ORDINATES ARE IN METRES UNLESS NOTED OTHERWISE.

2. INVERT LEVEL (IL) HAVE BEEN INDICATED FOR POINT OF SEWERAGE DRAIN AND UNIFORM SLOPE SHOULD BE PROVIDED BETWEEN THE MANHOLE.

3. MANHOLE LOCATIONS NEAR TOILET ARE TENTATIVE.

4. LENGTH OF SEWER WATER PIPE BETWEEN MANHOLES ARE INDICATED IN THIS DRAWING. THESE LENGTHS ARE APPROXIMATE (HORIZONTAL) AND TO BE ADJUSTED DURING CONSTRUCTION FOR PROPER FIXING.

5. SEWERAGE COLLECTION PIPE/DRAIN SHALL BE PROVIDED FROM TOILET/OTHER FACILITIES LOCATIONS UP TO MANHOLES WHEREVER REQUIRED. OTHERWISE CONNECTION PIPE/DRAIN SHALL BE PROVIDED BETWEEN S.P/W.P. AND NEAREST SEWERAGE MANHOLE.

6. THIS DRAWING IS APPLICABLE ONLY FOR SEWERAGE DRAIN NETWORK AND MANHOLE LOCATIONS. ALL OTHER INFORMATION, eg. LEVEL(EC.),FACILITY OUTLINE,GRADING ETC. ARE SHOWN IN THIS DRAWING FOR INFORMATION ONLY.

7. INVERT LEVELS (IL) HAVE BEEN INDICATED FOR THE MANHOLES AND UNIFORM SLOPE SHOULD BE PROVIDED IN THE SANITARY SEWER LINES BETWEEN THE MANHOLES.

8. PIPE BETWEEN LIFTING STATION (LS) AND NEXT MANHOLE SHALL BE PRESSURE PIPE, FOR LAYOUT AND DETAILS REFER UTILITY DRAWINGS.

9. FOR S.P./W.P. LOCATION AND DETAILS REFER ARCHITECTURAL DRAWINGS.

10. THIS DRAWING SHOULD BE READ IN CONJUNCTION WITH DRG. NO. MD-216-49EE-EG-CV-DED-1001, MD-216-49EE-EG-CV-DED-1002 & MD-216-49EE-EG-CV-DED-1004.

LIST OF DRAWING NO. OF LIFTING STATION :-

| LIFTING STATION | DRAWING NO. |
|---|---|
| LS-1 | MD-216-49EE-EG-CV-GAD-1001 |
| LS-2 | MD-216-49EE-EG-CV-GAD-1002 |
| LS-3 | MD-216-49EE-EG-CV-GAD-1003 |
| LS-4 | MD-216-49EE-EG-CV-GAD-1004 |
| LS-5 | MD-216-49EE-EG-CV-GAD-1005 |
| LS-6 | MD-216-49EE-EG-CV-GAD-1006 |

LS=LIFTING STATION, FSLP=FINAL SEWERAGE LIFTING PIT

| Schedule of Sewer Water PVC Pipes / Encased PVC Pipes | | |
|---|---|---|
| Pipes | Diameter | Slope |
| D1 | 150 | 1/100 |
| D2 | 200 | 1/150 |
| D3 | 250 | 1/150 |

| REV | DATE | REVISION DESCRIPTION | DRN | CHK | ENG | APP | EMP |
|---|---|---|---|---|---|---|---|
| D01 | 16/02/2023 | AS-BUILT | PML | BJB | APEX | SJM | |
| C05 | 08/02/2022 | ISSUED FOR CONSTRUCTION AND REVISED AS MARKED (05) | PML | BJB | APEX | SJM | |
| C04 | 05/05/2021 | ISSUED FOR CONSTRUCTION | APEX | BJB | APEX | SJM | |
| C03 | 19/04/2021 | ISSUED FOR CONSTRUCTION AND REVISED AS MARKED (03) | APEX | BJB | APEX | SJM | |
| C02 | 20/01/2021 | ISSUED FOR CONSTRUCTION AND REVISED AS MARKED (02) | APEX | BJB | APEX | SJM | |
| C01 | 23/07/2020 | ISSUED FOR CONSTRUCTION AND REVISED AS MARKED (01) | APEX | MLI | APEX | SJM | |
| A01 | 23/06/2020 | ISSUED FOR REFERENCE | APEX | MLI | APEX | SJM | |

| DOCUMENT NUMBER | REFERENCE DOCUMENTS |
|---|---|
| MD-216-1000-EG-GE-PPD-1002 | PROCESS PLANT LAYOUT |

RESERVER FOR VENDOR LOGO AND DRAWING NUMBER

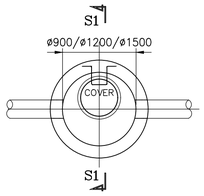
LARSEN & TOUBRO DRAWING NUMBER: 019045_910_C_V_24021_003 — REVISION: 6
OUTOTEC DRAWING NUMBER: 900707__GAU01__AG90__CTB17__50002 — DOCUMENT ID: OU100673152 — REVISION: 6

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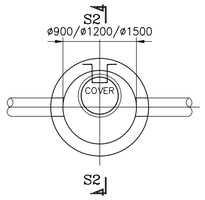
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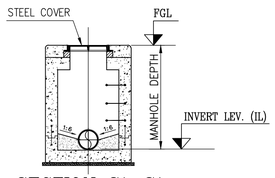
| | | | |
|---|---|---|---|
| DRAWN | APEX | 19/06/2020 | MANSOURAH - MASSARAH GOLD PROJECT |
| CHECKED | SNM | 22/06/2020 | DETAIL DRAWING |
| ENGINEER | APEX | 17/06/2020 | PROCESS PLANT SEWERAGE NETWORK |
| APPROVED | SJM | 23/06/2020 | LAY. & DETAILS OF SEWERAGE NETWORK SH.-1 |
| EMPLOYER | | | |
| SCALE 1:500 | | A1 | DRAWING NUMBER: MD-216-49EE-EG-CV-DED-1003 — REV: D01 |



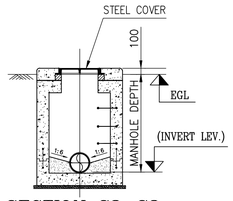

PLAN OF MANHOLE PIT
(CIRCULAR TYPE )



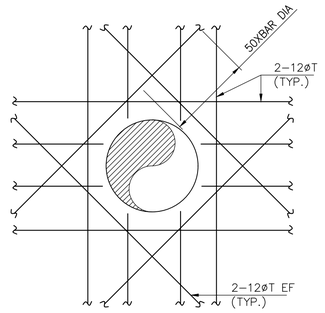

PLAN OF MANHOLE PIT
(CIRCULAR TYPE )



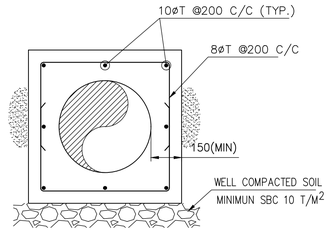

SECTION S1–S1
AT PAVEMENT / FGL AREA (TYPE-1)
SCALE=1:50



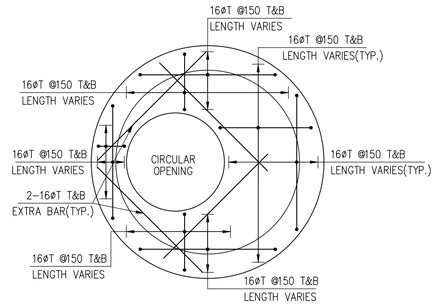

SECTION S2–S2
AT OPEN / EGL AREA (TYPE-2)
SCALE=1:50



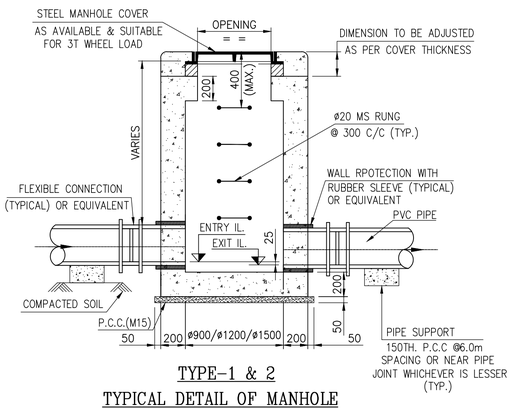

TYP. REINF. DETAIL
OF PIPE OPENING
SCALE=1:10



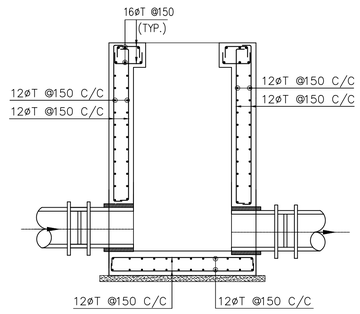

CONCRETE ENCASING AROUND PIPE(TYP.)
(BELOW ROAD & HEAVY VEHICULAR MOVEMENT AREA)
(UPTO 500mm BEYOND ROAD KERB BOTH SIDE)
SCALE=1:10



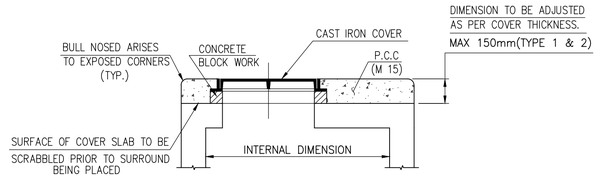

COVER SLAB FOR MANHOLE
(TYPE 1 & 2)
SCALE=1:25




TYPE–1 & 2
TYPICAL DETAIL OF MANHOLE
SCALE=1:25




TYPE–1 & 2
TYPICAL REINFORCEMENT DETAIL
SCALE=1:25




TYPICAL SECTION OF COVER OPENING
SCALE=1:25

## LEGENDS: –

LS = LIFTING STATION
FLSP = FINAL SEWERAGE LIFTING PIT
EGL = EXISTING GROUND LEVEL
FGL = FINISHED GROUND LEVEL
DRG. = DRAWING
S.P. = SOIL PIT
W.P. = WASTE PIT
LEV. = LEVEL
REINF.=REINFORCEMENT
T&B = TOP AND BOTTOM
TYP. = TYPICAL
P.C.C = PLAIN CEMENT CONCRETE
MAX. = MAXIMUM

## NOTES:~

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND INVERT LEVEL & CO–ORDINATES ARE IN METRES UNLESS NOTED OTHERWISE.

2. GRADE OF REINFORCED CEMENT CONCRETE SHALL BE M35. ENCASED CONCRETE GRADE IS M20

3. REINFORCEMENT BARS SHALL CONFORM TO ASTM A615 GRADE 60 (fy=420 N/sq.mm.)

4. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE 25mm(MINIMUM) U.N.O.

5. INVERT LEVEL (IL) HAVE BEEN INDICATED FOR POINT OF SEWERAGE DRAIN AND UNIFORM SLOPE SHOULD BE PROVIDED BETWEEN THE MANHOLE.

6. AT CONSTRUCTION JOINTS, WATER –TIGHTNESS SHALL BE ENSURED BY PROVISION OF SUITABLE RUBBER/ PVC WATER STOPS 150MM OR 250MM WIDE 'SIKA–WATERBARS–H' OR EQUIVALENT) OR ACRYLIC SWELLABLE SEALING PROFILE ( 'SIKA–SWELL A' OR EQUIVALENT).

7. MAXIMUM LOAD CARRYING CAPACITY MUST BE ENGRAVED ON MANHOLE COVER.

8. MANHOLE COVER SHELL BE AS PER BS EN 124.

9. THIS DRAWING SHOULD BE READ IN CONJUNCTION WITH DRG. NO. MD–216–49EE–EG–CV–DED–1001, MD–216–49EE–EG–CV–DED–1002 & MD–216–49EE–EG–CV–DED–1004.

| DOCUMENT NUMBER | REFERENCE DOCUMENTS | REV | DATE | REVISION DESCRIPTION | DRN | CHK | ENG | APP | EMP |
|---|---|---|---|---|---|---|---|---|---|
| | | D01 | 16/02/2023 | AS–BUILT | PML | BJB | APEX | SJM | |
| | | C05 | 08/02/2022 | ISSUED FOR CONSTRUCTION AND REVISED AS MARKED 6A | PML | BJB | APEX | SJM | |
| | | C04 | 25/05/2021 | ISSUED FOR CONSTRUCTION | APEX | BJB | APEX | SJM | |
| MD–216–49EE–EG–CV–DED–1003 | LAYOUT & DETAILS OF SEWERAGE NETWORK SH.–1 | C03 | 19/04/2021 | ISSUED FOR CONSTRUCTION | APEX | BJB | APEX | SJM | |
| MD–216–49EE–EG–CV–DED–1002 | LAYOUT & DETAILS OF SEWERAGE NETWORK | C02 | 20/01/2021 | ISSUED FOR CONSTRUCTION AND REVISED AS MARKED 6A | APEX | BJB | APEX | SJM | |
| MD–216–49EE–EG–CV–DED–1001 | LAYOUT & DETAILS OF SEWERAGE NETWORK | C01 | 23/07/2020 | ISSUED FOR CONSTRUCTION AND REVISED AS MARKED 6A | APEX | SNM | APEX | SJM | |
| MD–216–1000–EG–GE–PPD–1002 | PROCESS PLANT LAYOUT | A01 | 23/06/2020 | ISSUED FOR REFERENCE | APEX | SNM | APEX | SJM | |

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LARSEN & TOUBRO DRAWING NUMBER: 019045_910_C_V_24021_003 — REVISION: 6
OUTOTEC DRAWING NUMBER: 900707__GAU01__AG90__CTB17__50002 — DOCUMENT ID: OU100673152 — REVISION: 6

| | | |
|---|---|---|
| DRAWN | APEX | 19/06/2020 |
| CHECKED | SNM | 22/06/2020 |
| ENGINEER | APEX | 17/06/2020 |
| APPROVED | SJM | 23/06/2020 |
| EMPLOYER | | |
| SCALE | 1:1 | A1 |

MANSOURAH – MASSARAH GOLD PROJECT
DETAIL DRAWING
PROCESS PLANT SEWERAGE NETWORK
LAY. & DETAILS OF SEWERAGE NETWORK SH.–2

DRAWING NUMBER: MD–216–49EE–EG–CV–DED–1003 — REV: D01